

BUSINESS REQUIREMENTS DOCUMENT (BRD) & DATA FLOW DIAGRAMS (DFD)

Enterprise Training Management System (LMS) for Fresher Onboarding & Skilling

DOCUMENT CONTROL

Attribute	Details
Document Title	Business Requirements Document (BRD) & Data Flow Diagrams (DFD)
System Name	Enterprise Training Management System / Learning Management System (LMS)
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1. COVER PAGE & EXECUTIVE SUMMARY

1.1 Purpose

The purpose of this document is to establish the comprehensive **Business Requirements Document (BRD)** and **Data Flow Diagrams (DFD)** for the **Training Management System (TMS / LMS)**. This application is engineered to streamline, automate, and track the end-to-end lifecycle of fresher employee corporate training—from initial HR onboarding, account activation, multi-factor authentication (MFA), and batch allocation through daily attendance, interactive learning, coding/non-coding assignment evaluation, proctored assessments, progress tracking, mentor alignment, and completion reporting.

1.2 Business Problem

Organizations recruiting large cohorts of fresher employees encounter severe operational bottlenecks when relying on manual training management, spread-sheet based tracking, and fragmented communication tools: * **Manual Fresher Enrollment:** High risk of human error, delay in account creation, missing credentials, and manual distribution of onboarding links. * **Opaque Progress Tracking:** Inability for HR leadership and trainers to obtain real-time visibility into topic completion, video consumption, and practical skill acquisition. * **Unreliable Attendance Tracking:** Difficulty tracking day-to-day attendance across multiple streams, leading to untracked absenteeism and delayed intervention. *

Inefficient Assignment & Assessment Workflows: Manual distribution of assignments, lack of automated code execution engines for candidate submissions, and subjective, slow manual grading workflows. * **Decentralized Schedules:** Schedules managed via ad-hoc spreadsheets or emails, leading to confusion, missed training sessions, and scheduling overlaps. * **Delayed Communication & Absence Follow-Ups:** Lack of automated escalation workflows when freshers are absent consecutively, creating accountability gaps. * **Fragmented Stakeholder Dashboards:** Absence of role-tailored dashboards for Admins, Trainers, Trainees, Mentors, and Single Points of Contact (SPOCs). * **Manual & Time-Consuming Reporting:** Generating performance analytics, attendance reports, and audit trails requires days of manual compilation.

1.3 Expected Business Value

- **100% Onboarding & Workflow Automation:** Automated token-based activation, MFA security, and batch enrollment drastically cut administrative overhead by over 80%.
- **Real-Time Visibility & Analytics:** Instant dashboard metrics for HR and Trainers showing batch health, attendance rates, and progress metrics.
- **Automated Absence Escalation Protocol:** Integrated 3-Day SLA absence workflow (Day 1 email reminder → Day 2 reminder/escalation → Day 3 SPOC/HR formal escalation) ensuring strict attendance adherence.
- **Scalable Coding & Assessment Engine:** Built-in hidden test case code runner and proctored MCQ/Coding assessment modules eliminate manual evaluation delays.
- **Security & Compliance:** JWT-driven auth, MFA verification, RBAC (Role-Based Access Control), password policy enforcement, and audit logs ensure strict corporate governance.

2. SYSTEM ARCHITECTURE STACK

The system is constructed on a cloud-ready, decoupled full-stack web architecture:



3. BUSINESS OBJECTIVES & SCOPE

3.1 Business Objectives

1. **Automate Employee Onboarding:** Enable bulk CSV onboarding, auto-generated activation tokens, self-service password setup, and mandatory MFA binding.
2. **Centralize Training Data Repository:** Establish a single source of truth for Users, Companies, Batches, Courses, Course Days, Topics, Learning Materials, Assessments, and Attendance records.
3. **Automate Attendance & Escalation Workflows:** Provide daily attendance marking with automated Day 1, Day 2, and Day 3 multi-tiered email notification rules.
4. **Dynamic Course & Schedule Delivery:** Map courses to structured daily learning plans with integrated video links, dynamic schedule calendars (Monday–Friday), and progress updates.
5. **Standardized Assignment & Assessment Engine:** Support non-coding, case study, project, and automated coding assignments with hidden test case evaluation alongside proctored assessments.
6. **Provide Role-Tailored Dashboards:** Empower Admin, Trainer, Trainee, and Mentor roles with specialized, context-aware dashboards.
7. **Ensure Security & Governance:** Implement JWT authentication, MFA verification, password history policies, session tracking, and granular role permissions.

3.2 System Scope

In-Scope

- Admin management of users, companies, batches, courses, dynamic training plans, schedules, and global reports.
- Trainer management of batch sessions, trainee rosters, daily attendance, assignment creation/grading, assessment authoring, and performance feedback.
- Trainee experience: onboarding, MFA setup, daily schedule view, video/topic progress tracking, code submission & test execution, assessment taking (with proctoring timer), and mentor connectivity.
- Mentor access to assigned trainees, attendance summary, assessment scores, and mentorship notes.
- Automated email notifications for activation, password resets, attendance escalation, assignment deadlines, and test alerts.

Out-of-Scope (Phase 1)

- Third-party HRMS/Payroll sync (planned for Phase 2).
- Native mobile apps for iOS/Android (Web Responsive application used in Phase 1).

4. STAKEHOLDERS & RESPONSIBILITIES

Stakeholder Role	Description	Primary Responsibilities
Admin	Corporate System Administrator / L&D Head	Manages master data (Companies, Users, Batches, Courses), assigns Trainers to batches, configures system parameters, monitors global dashboards, and reviews company-wide reports.

Stakeholder Role	Description	Primary Responsibilities
Trainer	Technical Subject Matter Expert / Instructor	Conducts daily sessions, updates training schedules, marks attendance, authors assignments & assessments, evaluates submissions, and reviews trainee progress.
Trainee / Fresher	Newly hired fresher employee undergoing training	Attends live/daily training, completes daily topics & videos, submits coding/non-coding assignments, takes timed assessments, and tracks personal progress.
Mentor	Senior Employee / Technical Guide assigned to trainee	Reviews assigned trainee progress, monitors attendance & assessment scores, provides guidance, and records mentorship feedback.
Company / HR	Corporate Client / Human Resources Department	Registers cohorts/freshers into the platform, monitors aggregate batch completion metrics, and receives Day 3 absence escalations.
SPOC (Single Point of Contact)	Designated Operations / Training Coordinator	Serves as the primary operational contact between HR and Trainers; manages attendance overrides and receives attendance escalations.

5. USER ROLES & PERMISSIONS MATRIX

Feature / Module	Admin	Trainer	Trainee	Mentor	Company / HR	SPOC
User Login & MFA Setup	C/R/U/D	C/R/U/D	C/R/U/D	C/R/U/D	C/R/U/D	C/R/U/D
Fresher Registration & Onboarding	C/R/U/D	R	-	-	C/R	R
Bulk User Import (CSV)	C/R/U/D	-	-	-	C/R	-
Course & Topic Management	C/R/U/D	C/R/U	R	R	R	R
Batch Management	C/R/U/D	R/U	R	R	R	R
Schedule Management	C/R/U/D	C/R/U/D	R	R	R	R
Mark Daily Attendance	C/R/U/D	C/R/U	-	-	-	R/U
Attendance Escalation Review	C/R/U/D	R/U	-	R	R	C/R/U
Assignment Authoring	C/R/U/D	C/R/U/D	R	R	R	R
Assignment Submission & Code Execution	-	-	C/R/U	-	-	-
Assignment Evaluation / Grading	C/R/U/D	C/R/U/D	R (Self)	R	R	R
Assessment Authoring & Question Bank	C/R/U/D	C/R/U/D	-	-	-	-
Assessment Attempt & Proctoring	-	R (Results)	C/R (Take)	R (Results)	R (Results)	R (Results)

Feature / Module	Admin	Trainer	Trainee	Mentor	Company / HR	SPOC
Progress Tracking & Analytics	C/R/U/D	C/R/U	C/R (Self)	C/R (Assigned)	R	R
Mentor Assignment & Connections	C/R/U/D	R/U	R	C/R/U	R	R
System Reports & Audit Logs	C/R/U/D	R (Batch)	R (Self)	R (Assigned)	R	R

Legend: C = Create, R = Read, U = Update, D = Delete, - = No Access

6. DETAILED FUNCTIONAL REQUIREMENTS

Module A: Employee Registration & Onboarding

- **FR-ONB-01:** Admin or Company HR shall register freshers individually or via bulk CSV upload containing First Name, Last Name, Email, Corporate Phone, Company ID, and Batch ID.
- **FR-ONB-02:** Upon registration, the system shall generate a cryptographically secure 256-bit `activation_token` with a 48-hour expiration period.
- **FR-ONB-03:** An automated HTML activation email containing a personalized verification link shall be dispatched to the candidate's registered email via SMTP.
- **FR-ONB-04:** The candidate clicks the activation link, validating the token. If valid, the candidate is prompted to create a password compliant with corporate security rules.
- **FR-ONB-05:** Upon password creation, account status transitions from `PENDING_ACTIVATION` to `MFA_REQUIRED`.

Module B: Authentication & Security

- **FR-AUTH-01:** The system shall authenticate users using email and password via HTTP POST `/api/v1/auth/login`.
- **FR-AUTH-02:** Upon credential validation, if MFA is enabled, the system demands an TOTP 6-digit code.
- **FR-AUTH-03:** Upon successful authentication, FastAPI generates a signed JWT Access Token (short-lived, e.g., 60 mins) and Refresh Token (long-lived, e.g., 7 days) containing User ID, Role, Batch ID, and Permissions.
- **FR-AUTH-04:** Enforce Password Policy: Minimum 8 characters, 1 uppercase, 1 lowercase, 1 number, and 1 special character. Reject previous 3 historical passwords checked against `password_history`.
- **FR-AUTH-05:** Self-service Password Reset via email token (`password_reset_tokens`) expiring in 15 minutes.
- **FR-AUTH-06:** Audit Logging: Maintain `login_history` storing IP address, user agent, login timestamp, status (SUCCESS/FAILED), and failure reason.

Module C: Dashboards

- **FR-DASH-01 Admin Dashboard:** Real-time counter metrics for Total Freshers, Active Batches, Active Courses, System Attendance %, At-Risk Trainees, and Pending Escalations.
- **FR-DASH-02 Trainer Dashboard:** Batch roster list, Today's Live Schedule, Pending Assignment Submissions requiring grading, Upcoming Assessments, and Attendance percentage gauge.
- **FR-DASH-03 Trainee Dashboard:** Personal Training Progress %, Monday-Friday Schedule Calendar, Ongoing Course Day & Topics, Assigned Assignments (Coding/Non-Coding) with due dates, Upcoming Proctored Tests, and

Notification Feed.

- **FR-DASH-04 Mentor Dashboard:** Assigned Trainees summary cards, Trainee Progress %, Attendance records, Assessment scores, and feedback entry box.

Module D: Course & Training Plan Management

- **FR-CRS-01:** System shall allow Admins and Trainers to create Courses structured hierarchy: `Course` → `CourseDay` → `CourseTopic` → `LearningMaterial` / `Video` .
- **FR-CRS-02:** Ability to attach video URLs (MP4/HLS streaming), PDF documents, coding exercises, and reference notes to specific course topics.
- **FR-CRS-03:** Assign Courses to specific Batches via `training_plans` , setting Start Date, Target End Date, and Active Status.
- **FR-CRS-04 Dynamic Training Plan Mapping:** Automatically shift course days based on organizational holidays and custom batch schedules.

Module E: Schedule Management

- **FR-SCH-01:** Trainers and Admins can create training sessions associated with a Batch, Course, and Trainer for specific dates and time slots.
- **FR-SCH-02:** Display interactive Monday-Friday weekly calendar view filtering by Batch, Course, and Trainer.
- **FR-SCH-03:** Status management for sessions: `SCHEDULED` , `IN_PROGRESS` , `COMPLETED` , `CANCELLED` , or `RESCHEDULED` .

Module F: Attendance Management & Automated Escalation Workflow

- **FR-ATT-01:** Trainers mark attendance per training session for all enrolled batch trainees with status: `PRESENT` , `ABSENT` , `LATE` , or `EXCUSED` .
- **FR-ATT-02 Automated 3-Day Absence Escalation Workflow:**
- **Day 1 Absence:** System detects trainee marked `ABSENT` . Automatically sends an automated **Day 1 Reminder Email** to the trainee urging them to attend the next session and review missed materials.
- **Day 2 Consecutive Absence:** System detects 2 consecutive days of absence. Sends a **Day 2 Warning & Escalation Email** to the Trainee and CCs the Trainer and SPOC.
- **Day 3 Consecutive Absence:** System detects 3 consecutive days of absence. Triggers a **Day 3 Critical Escalation Notice** dispatched directly to HR, SPOC, Trainer, and Admin. Trainee status is flagged as `AT_RISK_ABSENCE` requiring manual SPOC/HR intervention to unlock further assessments.

Module G: Assignment Management

- **FR-ASG-01:** Support 4 assignment types: `CODING` , `NON_CODING` , `PROJECT` , and `CASE_STUDY` .
- **FR-ASG-02 Coding Assignments:**
- Map to `coding_problems` . Display problem statement, constraints, input/output format, sample test cases, and online code editor.
- Execute code against hidden test cases (`hidden_test_cases`) via background evaluation process.
- Store exact code submission, execution output, memory usage, runtime duration, and pass/fail score in `coding_submissions` .
- **FR-ASG-03 Non-Coding / Case Study / Project Assignments:**

- Provide downloadable instruction PDFs and submission upload portal (PDF, ZIP, DOCX up to 25MB).
- Trainers grade uploaded files manually, entering scores, feedback text, and rubric ratings into `assignment_submissions`.

Module H: Assessment Management

- **FR-ASM-01:** Support two assessment modes: `MCQ` (Multiple Choice Questions) and `CODING`.
- **FR-ASM-02:** Assessments must originate from `assessments` table and draw total duration from `duration_minutes`.
- **FR-ASM-03:** Enforce single or multi-attempt rules using `assessment_attempts` table.
- **FR-ASM-04:** Coding questions within assessments map through `assessment_problems` to `coding_problems`.
- **FR-ASM-05:** Calculate and store total score, pass/fail status, answer sheet (`assessment_answers`), and attempt timestamps.

Module I: Proctored Test Engine

- **FR-PRC-01 Verification:** Trainee confirms identity prior to starting a proctored assessment attempt.
- **FR-PRC-02 Timer & Navigation:** Client-side countdown synced with backend server clock (`start_time` + `duration_minutes`). Allow sequential or free question navigation.
- **FR-PRC-03 Proctoring Controls:** Detect tab switching, window blur events, full-screen exits, and copy-paste attempts. Log violations to assessment attempt audit. Auto-submit if violation threshold (e.g. 3 tab switches) is breached.
- **FR-PRC-04 Auto-Submission:** When the server timer reaches 00:00:00, the system automatically collects, evaluates, and locks the attempt.
- **FR-PRC-05 Naming Convention:** Derived from `course_day_topic` (e.g. "Day 05 - Python Async & ORM Assessment").

Module J: Progress Tracking

- **FR-PRG-01:** Calculate real-time overall training progress percentage per trainee using weighted formula:

$$\text{Progress} = (0.20 \times \text{Attendance}\%) + (0.30 \times \text{Video/Topic}\%) + (0.25 \times \text{Assignments}\%) + (0.25 \times \text{Assessments}\%)$$
- **FR-PRG-02 Status Transitions:** Automatically update training status: `NOT_STARTED`, `IN_PROGRESS`, `COMPLETED`, or `FAILED`.

Module K: Trainer Module

- **FR-TRN-01:** Access batch roster, daily lesson plan, session scheduler, live attendance marking console, assignment grading portal, and trainee performance matrix.

Module L: Admin Module

- **FR-ADM-01:** Master user creation, company tenant setup, batch scheduling, trainer assignment, system config, global CSV imports, and audit log inspection.

Module M: Mentor Module

- **FR-MNT-01:** View assigned mentees, track mentee daily attendance, inspect assignment/assessment scores, schedule 1-on-1 check-ins, and record mentorship notes.

Module N: Notifications Engine

- **FR-NOT-01:** Multi-channel notification pipeline (Email via SMTP + In-App Notifications table) supporting Account Activation, Password Reset, Attendance Day 1/2/3 Escalation, Assignment Published, Assessment Due, and Schedule Changed alerts.
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7. NON-FUNCTIONAL REQUIREMENTS (NFR)

7.1 Performance Requirements

- **NFR-PER-01 Response Time:** 95% of standard REST API endpoints (FastAPI) must respond in under 200ms under normal load.
- **NFR-PER-02 Query Optimization:** Zero N+1 query patterns; mandatory use of SQLAlchemy `selectinload` / `joinedload` for relational fetching.
- **NFR-PER-03 Database Connection Pooling:** Maintain async database pool (min 10, max 50 connections) with connection health checks.

7.2 Security Requirements

- **NFR-SEC-01 Encryption:** Passwords hashed using `bcrypt` (work factor 12). All HTTPS communications enforced via TLS 1.3.
- **NFR-SEC-02 Authorization:** Strict Role-Based Access Control (RBAC) enforced via FastAPI dependency injection guards on every endpoint.
- **NFR-SEC-03 Input Sanitization & SQLi Protection:** Async SQLAlchemy ORM parameterized queries to prevent SQL injection. Pydantic schema validation for strict payload validation.
- **NFR-SEC-04 File Upload Security:** Restrict file extensions (PDF, DOCX, ZIP, PY, JS, JAVA, CPP), validate MIME types, and limit max file size to 25MB.

7.3 Scalability & Availability

- **NFR-SCA-01 Multi-Tenancy & Capacity:** Architected to support 10,000+ active trainees across 500+ concurrent batches without performance degradation.
- **NFR-AVL-01 System Uptime:** 99.9% uptime during operational business hours (08:00 AM - 08:00 PM).

7.4 Maintainability & Usability

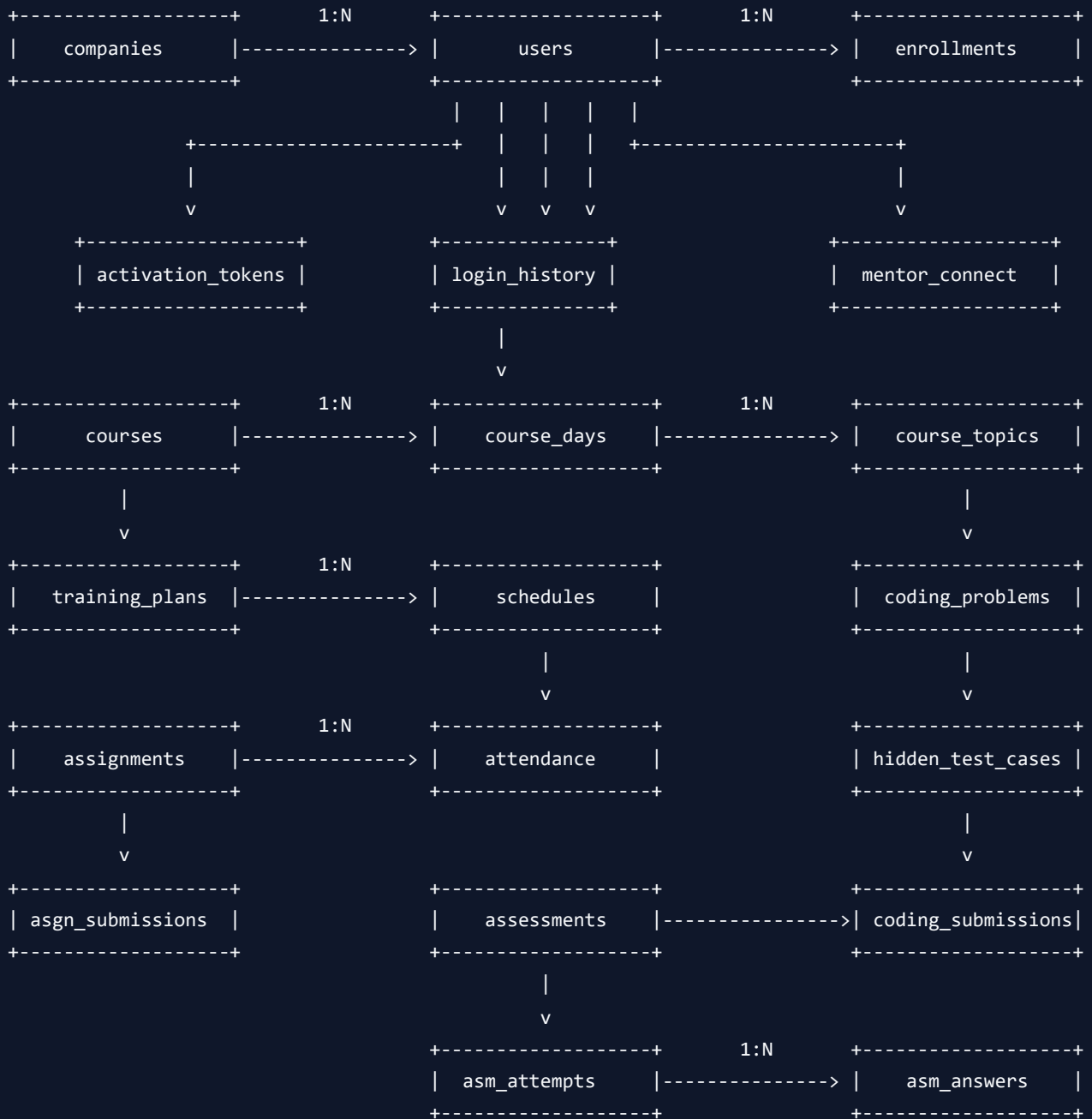
- **NFR-MNT-01 Database Migrations:** All schema changes executed cleanly via Alembic migration scripts.
 - **NFR-USA-01 Responsive UX:** Modern React UI supporting mobile, tablet, and high-density desktop displays with accessible contrast ratios.
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8. BUSINESS RULES SPECIFICATION

Rule ID	Business Rule	Module	Description	Trigger	Expected Result
BR-001	Account Activation Expiry	Onboarding	Activation tokens expire exactly 48 hours after generation.	User clicks activation link after 48h.	Token rejected; user prompted to request a new activation link.
BR-002	Mandatory MFA Setup	Security	Newly activated users must configure MFA prior to full application access.	First login post activation.	User redirected to /setup-mfa QR binding screen.
BR-003	Day 1 Absence Notification	Attendance	Trainee marked ABSENT on Day 1 receives an automated reminder email.	Trainer submits Day 1 attendance as ABSENT .	System generates Day 1 reminder email to Trainee.
BR-004	Day 2 Absence Escalation	Attendance	Trainee marked ABSENT 2 consecutive days triggers escalation warning.	Trainer submits Day 2 consecutive ABSENT .	System sends warning email to Trainee, Trainer, and SPOC.
BR-005	Day 3 Absence Critical Escalation	Attendance	Trainee marked ABSENT 3 consecutive days triggers HR escalation and status freeze.	Trainer submits Day 3 consecutive ABSENT .	System sends Critical Notice to HR/SPOC and sets status to AT_RISK_ABSENCE .
BR-006	Hidden Test Case Execution	Assignments	Coding assignment submissions must pass 100% of hidden test cases for max score.	Trainee submits code.	Code execution engine evaluates against hidden_test_cases and returns pass percentage.
BR-007	Strict Assessment Duration	Assessments	Candidate cannot submit or modify answers once duration_minutes expires.	Server clock reaches start_time + duration_minutes .	Server forces submission of current assessment_answers and locks attempt.
BR-008	Max Assessment Attempt Enforcer	Assessments	Candidate cannot exceed maximum attempts configured in assessments.max_attempts .	Trainee clicks "Start Assessment".	If attempts_count >= max_attempts , access is denied.
BR-009	Course Day Dependency	Training Plan	Trainee cannot access Day \$N+1\$ content until Day \$N\$ core videos are completed.	Trainee attempts to open Day \$N+1\$.	Topic locked unless Day \$N\$ video progress \$= 100\%\$.
BR-010	Dynamic Progress Calculation	Progress	Progress updates automatically upon attendance submission, video completion, or assignment/test grading.	DB insert/update on attendance, submission, or attempt.	Trainee overall progress recalculates and updates progress table.

9. DATA REQUIREMENTS & ENTITY SCHEMAS

The system database consists of 25 relational tables in PostgreSQL managed via SQLAlchemy ORM and Alembic:

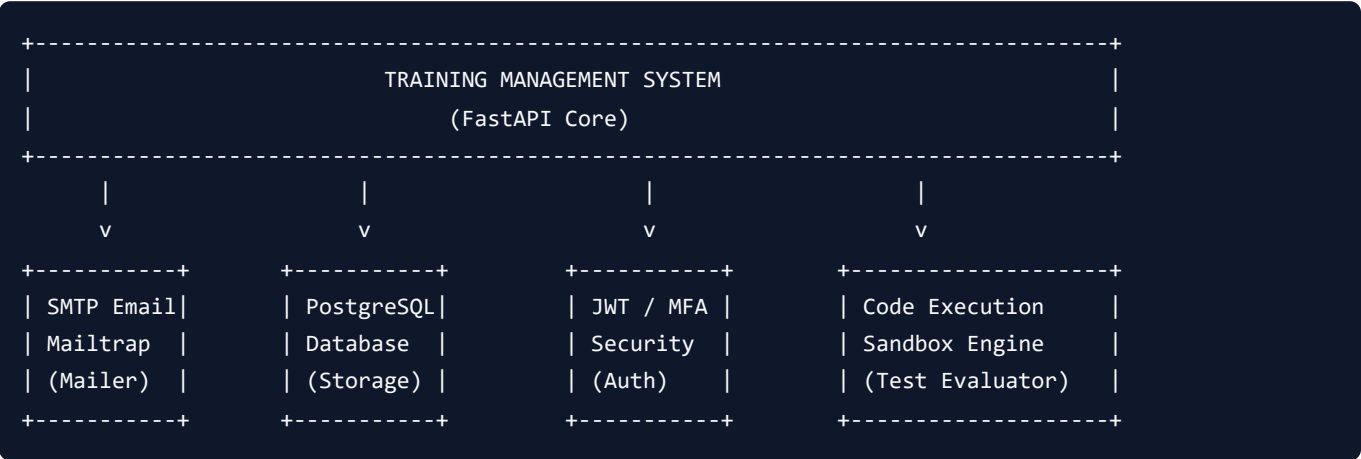


Key Data Entities & Descriptions

- users** : Master store for user identity, email, hashed password, role (`ADMIN` , `TRAINER` , `TRAINEE` , `MENTOR` , `SPOC` , `COMPANY_HR`), status, MFA secret, phone.
- activation_tokens** : Tokens for initial account activation, user link, token hash, created time, expires at, used status.
- password_reset_tokens** : Reset tokens, user link, token hash, expires at, used flag.
- login_history** : Audit record of login attempts, IP address, user agent, success flag, failure reason, timestamp.
- companies** : Corporate client profiles, tenant name, corporate email domain, contact details.

6. `batches` : Cohort groupings of freshers, batch code, start date, end date, company ID, designated trainer ID.
 7. `courses` : Master course catalog, title, description, technology stack tag, total days count.
 8. `course_days` : Breakdown of course into individual days (Day 1, Day 2...), day title, summary.
 9. `course_topics` : Granular learning units per day, topic name, video URL, duration, reading material link.
 10. `training_plans` : Mapping between Courses and Batches, specifying custom start/end dates and schedule configuration.
 11. `enrollments` : Association table mapping Trainee Users to specific Batches and Courses with status tracking.
 12. `schedules` : Calendar event entries for live sessions, batch ID, course day ID, trainer ID, start time, end time, session status.
 13. `attendance` : Daily attendance logs per trainee, schedule ID, trainee ID, status (`PRESENT` , `ABSENT` , `LATE` , `EXCUSED`), marked by trainer ID.
 14. `attendance_followups` : Tracking table for automated absence escalations (Day 1, Day 2, Day 3 status, sent timestamps, notes).
 15. `assignments` : Problem descriptions assigned to trainees, title, type (`CODING` , `NON_CODING` , `PROJECT` , `CASE_STUDY`), course day ID, max score, deadline.
 16. `assignment_submissions` : Trainee submissions for non-coding/project assignments, file URL, submission text, score, feedback, graded by trainer ID.
 17. `coding_problems` : Technical coding problem bank, title, statement, constraints, sample inputs/outputs, difficulty level.
 18. `hidden_test_cases` : Test cases used for automated evaluation, coding problem ID, input data, expected output data, points weight.
 19. `coding_submissions` : Code submitted by trainees, source code, language, problem ID, test cases passed, score, compiler output, execution time.
 20. `assessments` : Configured tests, title, type (`MCQ` , `CODING`), total duration in minutes, pass marks percentage, max attempts allowed.
 21. `assessment_problems` : Junction table mapping assessments to specific coding problems or MCQ questions with order and points.
 22. `assessment_attempts` : Record of a trainee starting an assessment, start time, end time, proctoring violation count, status, calculated score.
 23. `assessment_answers` : Specific candidate selections/answers for each question within an attempt.
 24. `progress` : Aggregated progress metric per trainee/course, attendance %, video %, assignment %, assessment %, overall status.
 25. `notifications` : Message queue store for in-app and email notifications, recipient user ID, title, body message, read flag, timestamp.
 26. `mentors` : Mentor profiles, employee code, domain expertise, max capacity.
 27. `mentor_connections` : Active pairing between a Mentor and a Trainee fresher with review notes and feedback.
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10. EXTERNAL SYSTEM INTEGRATIONS



1. **SMTP Email Service (Mailtrap / Enterprise Mailer):** Used to dispatch account activation emails, password reset tokens, Day 1/2/3 attendance notifications, assignment submission confirmations, and assessment alerts.
2. **PostgreSQL Relational Database:** Primary persistent data store managed asynchronously via SQLAlchemy Async Engine and Alembic for schema migration management.
3. **Authentication & Security Engine (PyOTP & PyJWT):** Handles TOTP QR code generation/validation for MFA and signs/verifies JWT tokens for API authorization.
4. **Code Execution / Evaluation Service:** Subprocess/Docker sandbox environment that executes candidate submitted Python/JS/Java code against `hidden_test_cases`, enforcing memory and CPU timeout restrictions.

11. END-TO-END BUSINESS WORKFLOWS

Workflow 1: Onboarding & Account Activation

Admin/HR Registers Candidate → Generate Token → Email Sent → Candidate Sets Password → MFA Bind → Login

1. HR/Admin inputs fresher details or uploads a bulk CSV file.
2. System creates `users` record with status `PENDING_ACTIVATION` and inserts `activation_token`.
3. System sends activation email with link `https://tms.company.com/activate?token=XYZ`.
4. Trainee clicks link; frontend calls API to validate token.
5. Trainee specifies a new password adhering to security policy.
6. Account status changes to `MFA_REQUIRED`. Trainee scans QR code using Google Authenticator and verifies 6-digit TOTP.
7. Account activated (`ACTIVE`). Trainee logs in and receives JWT auth tokens.

Workflow 2: Batch Allocation & Training Plan Setup

1. Admin creates a new `company` and `batch` (e.g., "Java-Fresher-2026-B1").
2. Admin selects a master `course` and creates a `training_plan` mapping the course to the batch.
3. Admin enrolls freshers into the batch (`enrollments`).
4. Trainer is assigned to the batch and training schedule is auto-populated across the calendar.

Workflow 3: Daily Training Session & Attendance Marking

1. Trainer logs in, accesses Dashboard, and opens today's session from `schedules` .
2. Trainer conducts live session and opens Attendance Console.
3. Trainer marks each trainee as `PRESENT` , `ABSENT` , `LATE` , or `EXCUSED` .
4. System commits records to `attendance` and triggers the Absence Processing Engine.

Workflow 4: Multi-Tiered Absence Escalation Protocol

1. **Day 1 Absence:** Trainee marked `ABSENT` . System creates record in `attendance_followups` and sends Day 1 Reminder Email to Trainee.
2. **Day 2 Absence:** If trainee is marked `ABSENT` on the next consecutive session, system updates follow-up status to `DAY_2_WARNED` and sends Warning Email to Trainee, Trainer, and SPOC.
3. **Day 3 Absence:** If trainee is marked `ABSENT` for 3 consecutive sessions, system updates status to `DAY_3_ESCALATED` , sets trainee status to `AT_RISK_ABSENCE` , sends Critical Notice to HR/SPOC, and alerts the assigned Mentor.

Workflow 5: Topic & Video Progress Update

1. Trainee opens Course Syllabus, selects current Day, and clicks a Topic video.
2. React frontend tracks video playback duration via video player events.
3. Upon reaching 100% video completion, API updates `video_progress` and recalculates overall `progress` .

Workflow 6: Assignment Submission & Code Execution

1. Trainee views assigned Coding Assignment on Dashboard.
2. Trainee writes code in web IDE and clicks "Run Tests".
3. Backend passes code to Sandbox Execution Engine; runs input against `hidden_test_cases` .
4. Engine compares actual stdout vs expected output, calculates score, and saves `coding_submissions` .
5. Trainee submits final solution; score is logged to `progress` .

Workflow 7: Proctored Assessment Execution

1. Trainee enters Assessment Portal; system verifies candidate identity and checks `assessment_attempts` .
2. Trainee clicks "Start Test". Server logs `start_time` and starts client countdown timer based on `duration_minutes` .
3. Trainee answers MCQs and writes code for coding problems. Tab switches are logged by proctoring script.
4. Upon timer expiry or manual submission, system evaluates answers, updates `assessment_attempts` , and displays result score.

Workflow 8: Mentorship Monitoring Workflow

1. Admin pairs Trainee with Mentor via `mentor_connections` .
2. Mentor logs into Mentor Dashboard and views list of assigned mentees.
3. Mentor inspects daily attendance rates, assignment scores, and test results.
4. Mentor enters monthly feedback notes and flags any required academic interventions.

Workflow 9: System Analytics & HR Reporting

- 1. Admin/HR selects Report Type: Attendance Summary, Batch Completion Rate, At-Risk Trainee Report, or Trainer Efficiency.
- 2. System queries aggregated views across attendance , progress , assessment_attempts , and assignments .
- 3. Reports rendered on screen with interactive charts and exportable as CSV/PDF files.

12. REPORTS & ANALYTICS SPECIFICATION

Report Name	Target Audience	Primary Metrics Included	Data Source Entities
Attendance Compliance Report	HR / Admin / SPOC	Trainee attendance %, Total Present/Absent days, Consecutively absent trainees, Escalation levels.	attendance , users , batches
Course Completion Report	Admin / Trainer	Batch progress %, Topic completion %, Video watch completion %, Target completion date variance.	progress , courses , enrollments
Assignment Performance Matrix	Trainer / SPOC	Assignment submission rates, Average scores per topic, Pass/Fail ratios, Hidden test case pass rates.	assignment_submissions , coding_submissions
Assessment Performance Analytics	Admin / Trainer	Mean score, Highest/Lowest scores, Pass %, Question discrimination index, Proctoring violation counts.	assessment_attempts , assessments
At-Risk Trainee Report	HR / SPOC / Mentor	Trainees with Progress < 60%, Attendance < 85%, or Day 3 Absence escalations.	progress , attendance , attendance_followups
Trainer Productivity Report	Admin / HR	Total sessions conducted, Grading turnaround time, Trainee feedback ratings per batch.	schedules , assignments , trainee_feedback

13. REQUIREMENTS TRACEABILITY MATRIX (RTM)

Req ID	Requirement Description	Target Module	User Role	Priority	Related DB Entity / API Endpoint	Status
RTM-01	Account activation via emailed token	Onboarding	Trainee	Critical	activation_tokens / /api/v1/auth/activate	Implemented
RTM-02	TOTP 2FA binding & verification	Security	All Roles	Critical	users.mfa_secret / /api/v1/auth/mfa/verify	Implemented

Req ID	Requirement Description	Target Module	User Role	Priority	Related DB Entity / API Endpoint	Status
RTM-03	Daily session attendance marking	Attendance	Trainer	Critical	attendance / /api/v1/attendance/mark	Implemented
RTM-04	3-Day Automated Absence Escalation	Attendance	System	Critical	attendance_followups / Cron Escalation Service	Implemented
RTM-05	Hidden test case code evaluation	Assignments	Trainee	Critical	hidden_test_cases , coding_submissions	Implemented
RTM-06	Timed proctored assessment execution	Assessments	Trainee	High	assessments , assessment_attempts	Implemented
RTM-07	Weekly Mon-Fri schedule calendar	Scheduling	All Roles	High	schedules / /api/v1/schedules/weekly	Implemented
RTM-08	Automated overall progress calculation	Progress	System	High	progress / Progress Engine Service	Implemented
RTM-09	Mentor progress monitoring dashboard	Mentorship	Mentor	Medium	mentor_connections / /api/v1/mentor/mentees	Implemented
RTM-10	CSV bulk user & batch enrollment	Onboarding	Admin	High	users , enrollments / /api/v1/admin/import-csv	Implemented

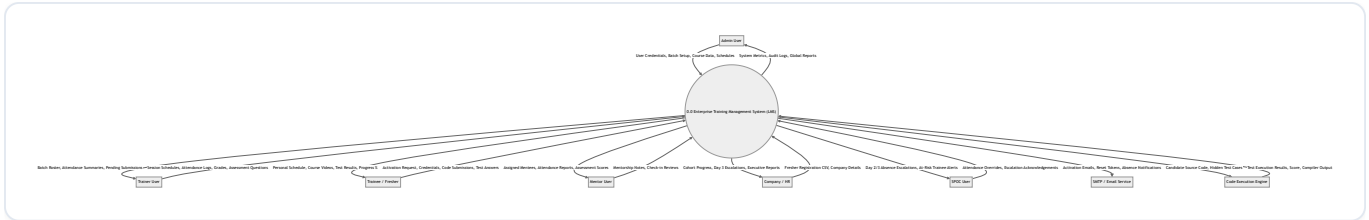
14. DATA FLOW DIAGRAMS (DFD)

DFD Notation Standards

- **External Entity:** Rectangular Box [Entity Name]
- **Process:** Rounded Box / Circle (P.N Process Name)
- **Data Store:** Parallel Open Block [(D.N Data Store Name)]
- **Data Flow:** Directed Arrow --> with labeled payload.

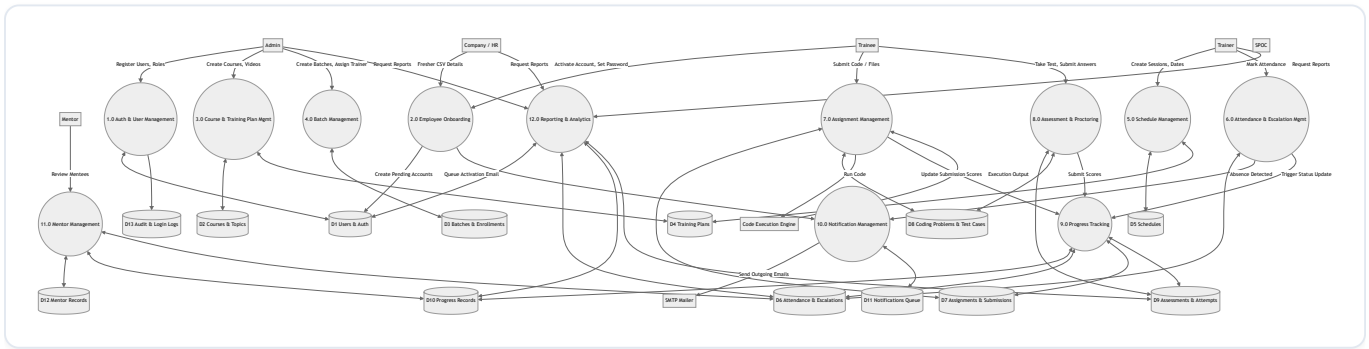
DFD LEVEL 0 – CONTEXT DIAGRAM

The Context Diagram presents the entire Training Management System as a single central process interacting with external system entities.



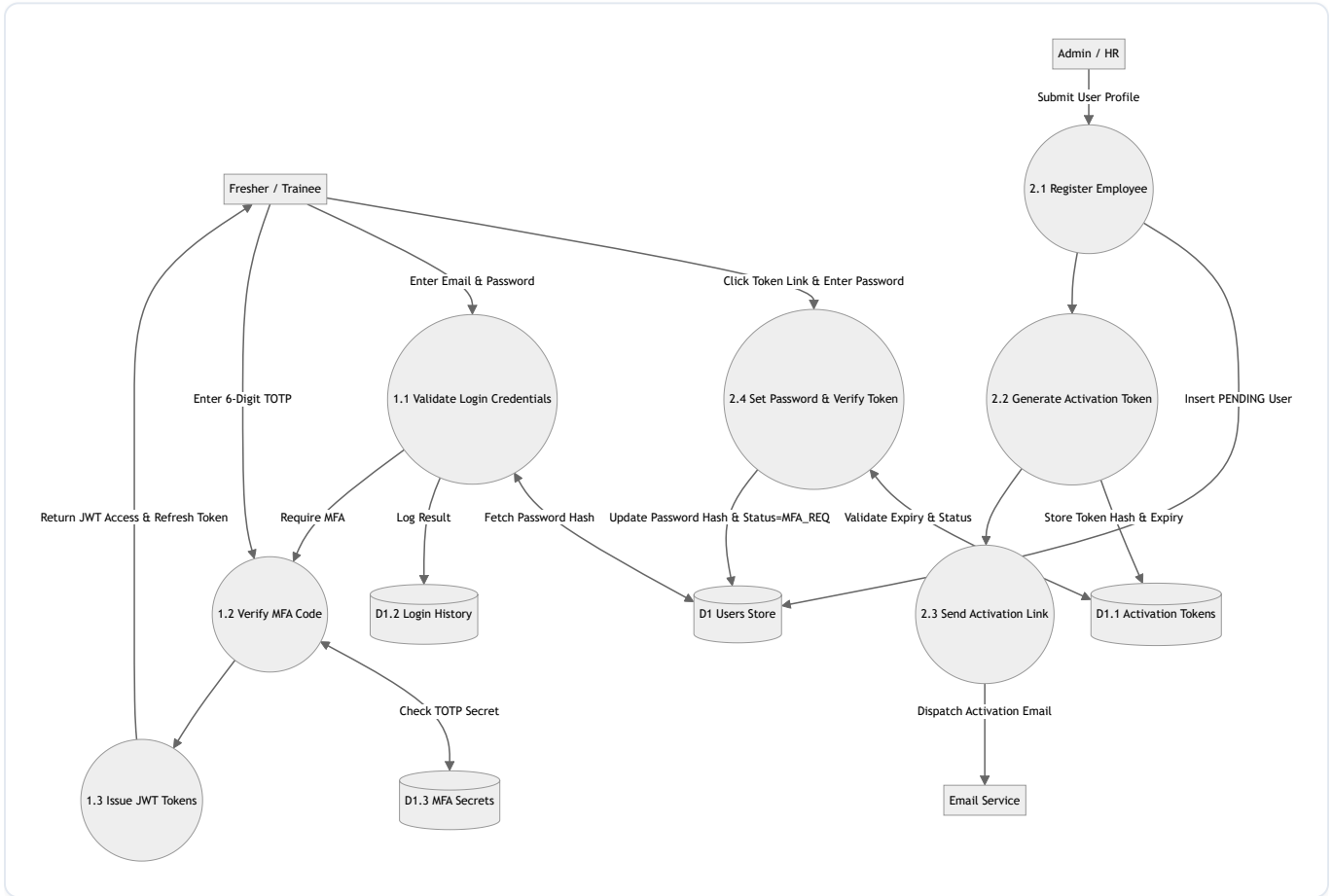
DFD LEVEL 1 – SYSTEM PROCESS DECOMPOSITION

Level 1 decomposes process **0.0** into 12 core sub-processes interacting with 13 key Data Stores.

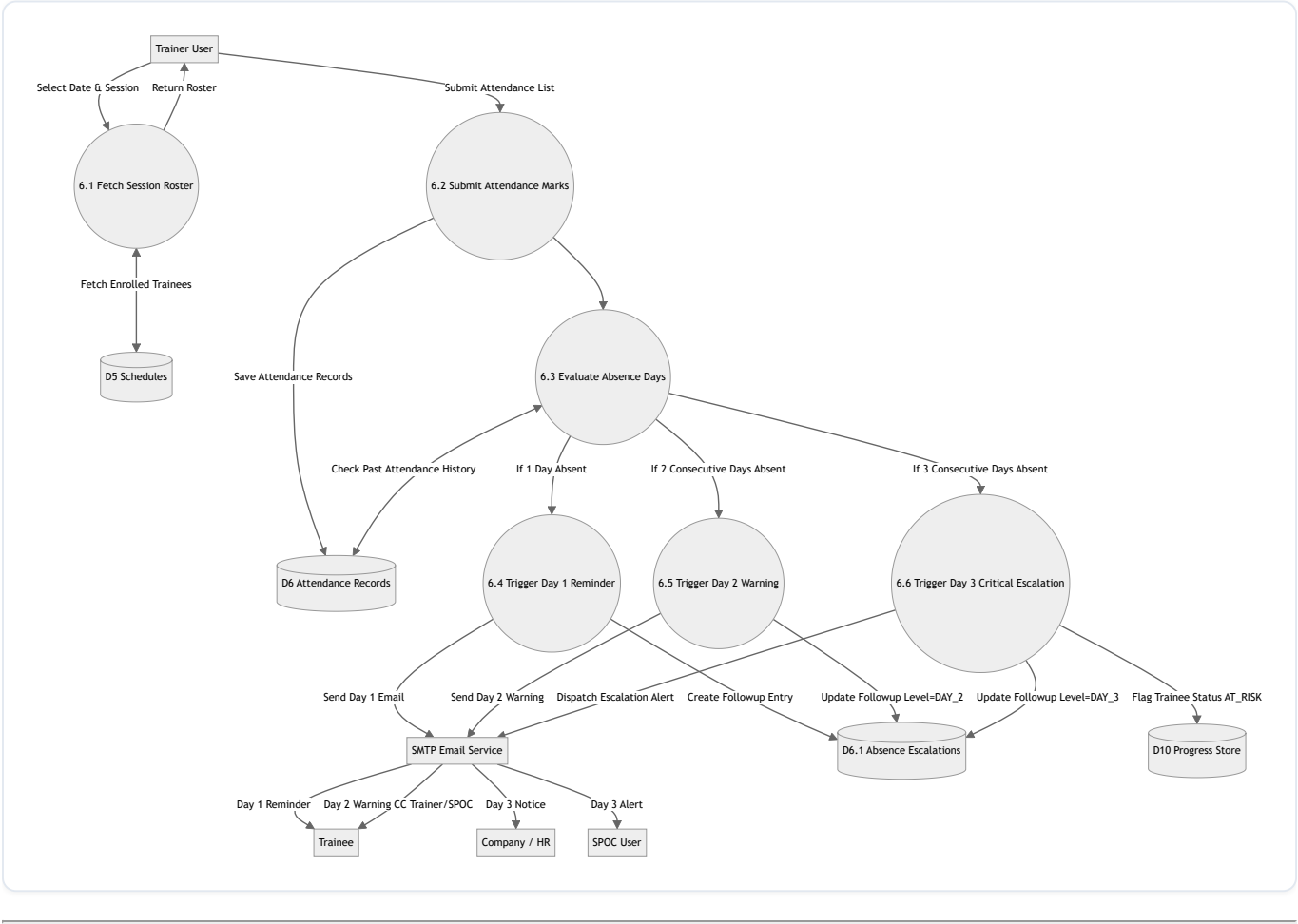


DFD LEVEL 2 – DETAILED PROCESS DIAGRAMS

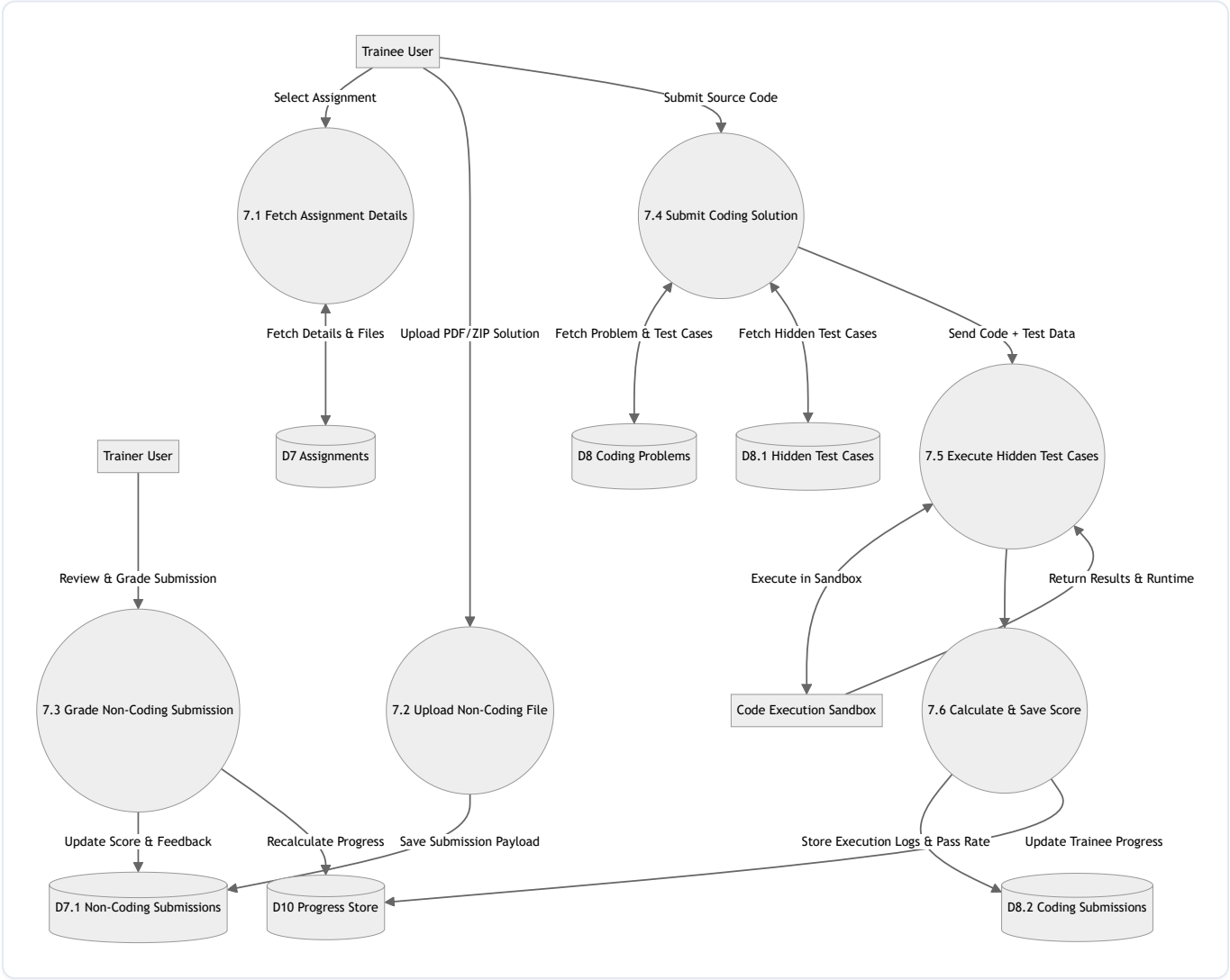
DFD Level 2 – Process 1.0 & 2.0: Authentication & Employee Onboarding



DFD Level 2 – Process 6.0: Attendance Management & Absence Escalation Workflow



DFD Level 2 – Process 7.0: Assignment Management (Coding & Non-Coding)



DFD Level 2 – Process 8.0: Assessment & Proctoring Controls



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DFD Level 2 – Process 9.0: Dynamic Progress Tracking Engine



15. ASSUMPTIONS, CONSTRAINTS & FUTURE ENHANCEMENTS

15.1 Assumptions

1. Corporate trainees possess standard desktop/laptop computers equipped with modern evergreen web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge).
2. SMTP credentials for Mailtrap (development) and corporate mail gateways (production) are configured and accessible with high deliverability rates.
3. Code execution sandbox environment maintains outbound network isolation to prevent executing malicious code against internal databases.

15.2 System Constraints

1. **Network Connectivity:** Trainees require active internet connection during training sessions and proctored assessments.
2. **File Size Limit:** Assignment and project attachment uploads are strictly capped at 25MB per file to optimize server storage.
3. **Assessment Time Constraints:** Server-side time enforcement will invalidate assessment submissions received after `start_time` + `duration_minutes` + 2 minutes grace buffer.

15.3 Future Enhancements (Phase 2 Roadmap)

1. **AI-Powered Code Analysis & Feedback:** Integration with LLM APIs to generate automated code refactoring feedback for freshers.
2. **Webcam Proctoring & Facial Recognition:** Automated AI facial detection during proctored tests to flag unauthorized secondary individuals.
3. **Third-Party HRMS Integration:** Bi-directional sync with Workday, SAP SuccessFactors, or Darwinbox for automatic employee master sync.
4. **Offline Video Sync Mobile App:** Native iOS and Android applications supporting offline viewing of encrypted training videos.

End of Business Requirements Document (BRD) and Data Flow Diagrams (DFD)